Junho Andrew Choi

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EDUCATION

University of California, Berkeley Expected May 2027
B.A. Data Science (Concentration: Computational Methods in Biology) and B.A. Integrative Biology *Berkeley, CA*
Coursework: Principles & Techniques of Data Science (Data 100), Data Structures (CS 61B), Probability for Data Science (Data C140), Optimization Modeling (IEOR 164), Ethics of Data (Data C104); **in progress:** Machine Learning (CS 189), Data Engineering (DS C101)

SKILLS

Languages: Python, SQL, Java, R
ML & AI: PyTorch, scikit-learn, NumPy, pandas, LLM evaluation, RAG, Text2SQL, knowledge graphs (Neo4j), SHAP, Grad-CAM, feature engineering
Data & Tools: PostgreSQL, Jupyter, Git, React, Vite, Figma, Photoshop
Spoken: Korean (native), English (fluent), Mandarin

EXPERIENCE

SmartMind AI May 2026 – Aug 2026
AI Researcher, Data Science Team *Seoul, Republic of Korea*
- Initiated and led a proof-of-concept NL2SQL pipeline for an enterprise ERP system serving ~1,300 companies, scoping a 15-question evaluation suite and mapping the 6 core tables required for full coverage
- Diagnosed 3 systemic accounting-logic defects causing syntactically valid LLM-generated SQL to return silently incorrect results, tracing a 40% discrepancy in a financial ratio to a single inconsistent join-reference pattern
- Redesigned evaluation from single-run pass/fail testing into a 43-run reproducibility framework (15 questions × 3 trials), converting an unverified 20/20 pass rate into a measured, defensible 13/15 benchmark
- Migrated ~15 business rules from a long-form prompt into structured Neo4j knowledge-graph nodes and debugged them against failure evidence, resolving all remaining structural defects

Asan Medical Center (AMC) May 2025 – Aug 2025
AI Research Intern *Seoul, Republic of Korea*
- Co-authored a research paper on a multimodal deep survival model for hepatocellular carcinoma, developing and fine-tuning 3D U-Net and ResNet3D architectures in PyTorch with GPU acceleration
- Built automated preprocessing pipelines for 3D CT imaging and clinical records, cutting manual data preparation time across modalities
- Conducted large-scale exploratory analysis for the VLM3D Multi-Abnormality Classification Challenge, informing the feature selection and strategy that placed 7th among multi-institutional teams
- Integrated SHAP and Grad-CAM interpretability to validate feature importance and communicate model reasoning to clinicians in high-stakes prediction settings

Invention Corps of Berkeley — Mozilla Firefox Aug 2025 – Feb 2026
Design Engineer Consultant *Berkeley, CA*
- Engineered a Firefox performance-monitoring extension in React and Vite that tracked, filtered, and sorted 50+ concurrent tabs by live memory and CPU usage
- Led survey-based user research with a 12-person team, translating findings into feature requirements and Figma interface designs that shaped the shipped extension

PROJECTS

- **rad-DINO — Radiology Representation Learning:** implemented and ran experiments with a self-supervised vision transformer for radiology imaging, evaluating learned representations for downstream classification in PyTorch
- **AI Tactical Assistant:** trained a PyTorch network to predict match outcomes from collected historical player and team data, paired with a PuLP linear-programming optimizer generating constrained optimal formations — used to win a 6v6 league
- **Activation Domain Prediction Pipeline:** developed a CNN-BiLSTM model classifying transcriptional activation domains from protein sequences using sliding-window feature engineering and K-means clustering, reaching F1 = 0.95 across 10 AD subclasses

LEADERSHIP & AWARDS

US 8th Army, 2ID HHBN Sep 2022 – Mar 2024
Non-Commissioned Officer *Republic of Korea*
- Led a 40+ soldier platoon as a bilingual Korean–English platoon leader, delivering ACFT training and military coursework to a 96% platoon pass rate
- Awarded the US Army Certificate of Achievement (Mar 2024) for leadership and bilingual communication